

Week 1 — Safety, Tools & Electrical Fundamentals

Sessions: 1 (Theory) · 2 (Lab) · **Total:** 4 hours

Primary LOs: LO-01, LO-02

Hands-on target: Session 2 ≈ 85% practical

Instructor Preparation Guide (Complete Day Before)

Pre-session setup

- [] Print student binders: Week 1 handouts, pre-assessment, lab sheets (labs/W01_*)
- [] Post shop safety rules (see handouts/safety_shop_poster.md)
- [] Verify eyewash, spill kit, Class C/ABC extinguisher accessible and tagged
- [] Stage PPE rack: glasses, face shields, Class 00 gloves + leather over-gloves, aprons
- [] Prepare 2–3 training boards OR de-energized carts labeled “TRAINING — VERIFY ZERO ENERGY”
- [] Set up 4 tool stations (see Session 2)
- [] Cal-check or battery-check each DMM; replace blown mA fuses as needed
- [] Prepare LOTO kits: locks, tags, hasps; one pack designated for LOTO drill
- [] Whiteboard: Ohm’s Law triangle; series vs parallel battery sketch ready
- [] Pre-assessment answer key ready (instructor only)

Carts / props / tools

Item	Qty	Notes
De-energized or training cart packs	2+	Main pack connector accessible
Digital multimeters (CAT III)	1 per 2 students	Fluke 87V or equiv.
Low-voltage trainers / 6V practice batteries	2–4	For series/parallel build
Assorted hand tools	Full set	At identification stations
Insulated screwdrivers & pliers	Demo set	Contrast with non-insulated
LOTO kits	1 per pair	
Jewelry bowl / “remove jewelry” sign	1	

Planted conditions (Session 2)

- One cable with intentional high resistance (corroded lug demo) — labeled after discovery
- One training board with open circuit for continuity hunt

- Do **not** plant live short hazards

Instructor mindset notes

- Zero tolerance for jewelry or skipped PPE on Day 1 — model the culture.
 - Praise “I don’t know / stop and ask” behavior publicly.
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Week 1 Learning Objectives

By end of Week 1, students will:

1. Demonstrate proper PPE selection for electrical and mechanical golf cart work.
 2. Perform Lockout/Tagout on a battery pack and verify zero energy with a meter.
 3. Explain Ohm’s Law and identify series vs parallel battery configurations (36V/48V).
 4. Safely measure DC voltage, resistance, and continuity with a DMM.
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Session 1 — Theory & Demonstration (2 hours)

Materials needed: Projector/whiteboard; labs/W01_preassessment.md; PPE samples; DMM for demo; sample 6V batteries or diagram; LOTO kit; handouts ohm_law_quick_reference.md, safety_shop_poster.md, recommended_tool_list.md; syllabus/expectations sheet.

Timed Agenda

Time	Min	Activity	Instructor Notes
0:00	5	Welcome, roster, facility map, emergency exits	Point out eyewash, extinguishers, first aid
0:05	10	Program overview, schedule, grading, certificate criteria	Emphasize $\geq 60\%$ hands-on; diagnose-before-replace
0:15	15	Pre-assessment (written, not graded for pass/fail)	10 short items — baseline only; collect quietly
0:30	5	Debrief expectations & “no stupid safety questions”	Normalize stopping work
0:35	15	Shop hazard ID: arc flash/high current, H ₂ explosion, acid, crush, pinch	Use photos if available; [DIAGRAM PLACEHOLDER: hazard poster collage — arc, swollen battery, crushed foot zone]
0:50	10	PPE selection walk-through	Pass around Class 00 gloves; explain leather over-gloves; steel-toe expectation
1:00	15	LOTO procedure lecture + demo on cart pack	Steps on whiteboard; one-hand rule; never assume pack is dead
1:15	5	Break / stretch	Keep students out of energized bays
1:20	20	Electrical fundamentals: V, I, R, Ohm’s Law, power	Work 3 example problems; series vs parallel for 36V (6×6V) & 48V
1:40	15	Multimeter orientation demo	Functions, CAT rating, probe discipline, never amps jack by accident
1:55	5	Preview Session 2 stations + assign pairs	Homework: skim Ohm’s Law handout

Key Teaching Points

- Golf cart packs are **high current** even at “only” 36/48V — arcs can melt tools and cause burns.
- Hydrogen from charging is explosive — ventilation is not optional.
- Series adds voltage; parallel adds capacity — wrong jumper = damage/fire risk.
- Meter in amps mode across a battery $\approx$ fused meter or worse.

Common Student Misconceptions

Misconception	Correction
"48V isn't dangerous like house voltage."	Current capability of packs is extreme; shorting with a wrench is a shop-ending event.
"If the key is off, the pack is safe."	Pack remains energized; LOTO the pack connector / main positive as procedure requires.
"Continuity beeper proves a cable can carry 300A."	Continuity $\neq$ low resistance under load; voltage drop testing comes later.
"Any meter is fine."	Use adequately rated CAT meters; damaged leads are unsafe.

Safety Reminders (Session 1)

- Remove rings, watches, steel bracelets before any bay work.
- Safety glasses required in shop from this session forward.
- Never work alone on pack-connected systems during training.

Instructor closing check

Confirm all students have PPE access plan for Session 2; collect pre-assessments.

Session 2 — Hands-On Lab (2 hours)

Materials needed: Tool stations, DMMs, training boards/carts, LOTO kits, labs/
W01_multimeter_scavenger.md, labs/W01_loto_competency.md, series/parallel kits, jewelry bowl.

Timed Agenda

Time	Min	Activity	Instructor Notes
0:00	5	Safety brief + jewelry check + PPE don	Instructor signs attendance only after PPE on
0:05	10	Station overview & rotation rules	4 stations × ~20 min; pairs stay together
0:15	20	Station A — Tool ID & inspection	Identify torque wrench, crimper, insulated tools; reject damaged tools
0:35	20	Station B — Multimeter scavenger	Measure bank V, cell/battery V, continuity hunt; complete lab sheet
0:55	5	Water / rotate buffer	
1:00	20	Station C — LOTO drill	Isolate pack, tag, verify zero energy at main terminals, instructor sign-off
1:20	20	Station D — Series/parallel build	Build 12V from 2×6V series; discuss 36/48V pack math; measure
1:40	10	Group debrief: wrong meter mode near-miss stories	Invite student observations
1:50	10	Cleanup, pack restore, competency stamps, exit ticket	Exit: “One hazard I will never ignore”

Key Teaching Points (Lab)

- Verify meter on known source before trusting a “0V” reading.
- Tag ownership: person who locks is person who removes (training simplification of full OSHA LOTO — teach principle).
- Document readings with units every time.

Common Lab Errors to Watch

- Probes in A jack while measuring voltage.
- Skipping zero-energy verify after disconnect.
- Calling a pack “dead” based on key switch alone.

Safety Reminders (Session 2)

- Face shield + gloves when breaking pack connections if corrosion/arc risk.
- No jewelry. One-hand probing when any residual voltage possible.
- Spill kit location reviewed before any battery handling.

Assessment Focus

- Safety quiz (end of week or start of Week 2) + practical LOTO/multimeter competency sign-off.
- Instructor observes safe meter use and PPE compliance continuously.

Week 1 Quiz (8 questions) — Administer end of Session 2 or start of Week 2

Name: __ Date: _____

Multiple Choice

1. Before working on a golf cart battery pack, the first electrical safety action is to:
A) Turn the key off and begin probing
B) Apply LOTO and verify zero energy with a meter
C) Remove the controller cover
D) Disconnect the charger only
2. Class 00 insulated gloves are primarily intended to protect against:
A) Hot exhaust only
B) Electrical shock on low-voltage systems when used as rated
C) Sulfuric acid only
D) UV from welding
3. In a series string of six 8V batteries, nominal pack voltage is approximately:
A) 8V
B) 24V
C) 48V
D) 96V
4. Which meter function is appropriate to check if a fuse is open (power removed)?
A) ACA
B) Continuity / resistance
C) Hz
D) Capacitance
5. Hydrogen gas hazard is greatest when:
A) Tires are inflated
B) Batteries are charging without ventilation
C) The cart is in reverse
D) The radio is on

Short Answer

1. State Ohm's Law in formula form and solve for current if $V = 48V$ and $R = 16\Omega$.
2. List three PPE items required for battery pack work in this program.
3. Why is "key off" not sufficient proof that a pack is safe to work on?

Answer Key (Instructor)

1. B

2. **B**
 3. **C** ($6 \times 8V = 48V$)
 4. **B**
 5. **B**
 6. **$V = I \times R$** (or $I = V/R$). **$I = 48/16 = 3A$**
 7. Accept any three: safety glasses, face shield, Class 00 gloves, leather over-gloves, apron, steel-toe boots
 8. Pack remains energized; key only opens control circuits — must isolate pack and meter-verify zero energy
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Competency Sign-Off Box (copy to student record)

Skill	Pass/Fail	Instructor Initials	Date
PPE selection & donning			
DMM voltage measurement			
DMM continuity/resistance			
LOTO + zero-energy verify			